SABA S KHAN

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EDUCATION

New York University, Tandon School of Engineering — **Brooklyn, NY**

Master of Science in Biomedical Engineering (BME), GPA 4.0 — May 2026

National GEM Consortium Fellow

Relevant Coursework: Biomedical Device Design and Development, Biomaterials, Biomedical Instrumentation, CAD for Biomedical Engineering, Anatomy & Physiology

University of Maryland, Baltimore County — **Baltimore, MD**

Bachelor of Science in Chemical Engineering, Biotechnology and Bioengineering Track, GPA 3.57 — May 2021

Golden Key International Honor Society, Dean's List — May 2020, Dec 2020, May 2021

TECHNICAL SKILLS

- **CAD & Fabrication:** SolidWorks, Fusion 360, AutoCAD, 3D Printing, Microfabrication, Photolithography, Laser Machining, Micro-CNC
- **Software & Data:** Python, SQL, MATLAB, Minitab, Simulink, Arduino, Smartsheet
- **Testing & Validation:** Sensor Integration, Fluidic & Pressure Testing, Prototype Troubleshooting, Validation Testing

PROFESSIONAL EXPERIENCE

Potomac Photonics Inc. — **Halethorpe, MD**

R&D Project Engineer – Technical Lead — Jan 2024 – Aug 2024

- Led cross-functional teams of 4-6 across 3 R&D programs while contributing to 5 biomedical technologies.
- Translated customer requirements into engineering plans and risk assessments for prototype design and validation.
- Standardized manufacturing workflows and trained technicians to transfer prototype processes into production.

R&D Project Engineer — Sept 2022 – Dec 2023

- Developed microfabrication processes and analyzed QA/QC data to troubleshoot manufacturing issues.
- Designed custom fixtures using Fusion360 and built pressure/fluidic test stations for device validation.

Maryland Department of the Environment (MDE) — **Baltimore, MD**

Regulatory and Compliance Engineer I – Biosolids Division — Jan 2022 – Sept 2022

- Reviewed engineering designs and documentation for regulatory compliance and technical requirements.

RESEARCH EXPERIENCE

New York University, Tandon School of Engineering — **Brooklyn, NY**

Graduate Research Engineer - Biomedical Device Development — Aug 2025 – May 2026

- Designed a multi-layer surrogate head model using Fusion360 and fabricated it via 3D printing, hydrogel casting, and intricate biomaterial selection to replicate skull physiological parameters and geometry.
- Integrated pressure sensors into the system to conduct blunt impact testing under elevated ICP conditions to assess the neuroprotective properties of pressurized CSF.
- Compiled technical documentation and presented findings weekly to the PI and external neurosurgery collaborators.

UMBC, Center for Advanced Sensor Technology (CAST) — **Halethorpe, MD**

Undergraduate Research Assistant — Jun 2019 – Aug 2020

- Engineered cellulose acetate microfluidic devices using laser machining and polymer microfabrication techniques to evaluate material suitability for rapid prototyping.

RELEVANT PROJECTS

New York University, Tandon School of Engineering — **Brooklyn, NY**

Patient–Specific Hip Implant Design from Femoral Geometry — May 2025

- Designed a custom hip implant in Fusion 360 from patient-specific 3D femoral anatomy.
- Generated 2D engineering drawings with strict GD&T and industry tolerance specifications.
- Optimized implant geometry for biocompatibility, load distribution, and reduced stress concentration.

ECG Signal Acquisition & Processing Circuit — Feb 2025 – May 2025

- Built hardware featuring an instrumentation amplifier, low-pass filter, and notch filter.
- Validated circuit performance using both simulated and live cardiac input data.